ANALYSIS REPORT

Phase 2 Harmonic And Kinematic PINN Campaign Results Report

Wave 5.2 Phase 2 Full-PINN Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes Wave 5.2 Phase 2 of the sixteen-phase full-PINN theory-validation roadmap. Phase 2 implemented and tested the repository's first target-free differentiable angular-oscillator residual, periodic value-and-slope closure, and an optional frozen Bauer analytical anchor.

The canonical campaign completed all eight direction-separated runs:

- campaign: `phase2_harmonic_kinematic_pinn_runtime_bounded_restart_2026_07_26` ;
- dataset: `polished_dataset` , causal `setpoints` input mode;
- common eligible split: `675` train, `194` validation, and `97` test conditions per direction;
- surfaces: `Fw` and `Bw` , never merged for selection;
- completed runs: `8` ;
- failed runs: `0` ;
- campaign duration: approximately `2 h 35 min` ;
- scalar program-best promotion: none;
- Phase 2 physics promotion: none.

  The phase is complete as a valid negative result. The infrastructure and falsification evidence advance to Phase 3, but no Phase 2 oscillator, periodic-boundary, or Bauer-anchor loss weight becomes a default physical ingredient.

Evidence Contract

The canonical restart used the exact Phase 1 paired-condition assignment signature:

```text
c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16
```

Three Phase 0 anomalous training conditions remained quarantined. Every arm used the same:

- causal angle, speed, torque, temperature, and direction contract;
- `675 / 194 / 97` direction-specific condition split;
- stride `8` ;
- batch size `4` ;
- distributed `4,096` -point curve cap;
- `64` physics collocation points per batch;
- `24` -epoch ceiling and patience `5` ;
- harmonic order set `1` , `3` , `39` , `40` , `78` , `81` , `156` , `162` , and `240` . The smallest audited source curve still supplied at least `1,350` angular samples after stride, more than `2.8` times the `480` -sample Nyquist minimum for order `240` . The training loop evaluated at least `10,800` physics collocation points per epoch. Earlier interrupted or diagnostic attempts are retained for provenance but are ineligible for this phase decision. Immutable candidate-registry snapshots bind the bounded curve-payload diagnostic to the eight canonical checkpoints.

Candidate Definitions

Arm	Architecture And Loss	Scientific Role
`H0`	Explicit Fourier heads, no physics loss	Structured data-driven control; not a full PINN
`H1`	Implicit harmonic heads plus normalized oscillator residual	Isolated governing-residual test
`H2`	`H1` plus periodic value and slope closure	Oscillator and kinematic-boundary test
`H3`	`H2` plus frozen Phase 1 Bauer coefficient anchor	Analytical-plus-PINN guidance test

For order \(k\), the target-free oscillator residual was:

```text
R_k = (1 / k^2) * d2 h_k / d theta2 + h_k
```

The normalization prevents high orders from dominating only because of their derivative scale. Periodic closure was evaluated at matched causal conditions:

```text
y(0, u) = y(2 pi, u) dy/dtheta(0, u) = dy/dtheta(2 pi, u)
```

Scalar Campaign Results

The campaign leaderboard is a training diagnostic only. It ranks all directions together by test MAE and therefore cannot make a program or physics promotion.

Rank	Arm	Surface	Test MAE [deg]	Test RMSE [deg]	Val MAE [deg]
1	`H0` Fourier control	`Fw`	0.001646	0.002040	0.001852
2	`H1` oscillator	`Bw`	0.001723	0.002210	0.001856
3	`H2` oscillator + periodic	`Fw`	0.001784	0.002245	0.002074

Rank	Arm	Surface	Test MAE [deg]	Test RMSE [deg]	Val MAE [deg]
4	`H3` oscillator + periodic + Bauer	`Bw`	0.001931	0.002469	0.002087
5	`H1` oscillator	`Fw`	0.001951	0.002401	0.002127
6	`H2` oscillator + periodic	`Bw`	0.001966	0.002545	0.002018
7	`H0` Fourier control	`Bw`	0.002066	0.002643	0.002077
8	`H3` oscillator + periodic + Bauer	`Fw`	0.002389	0.003042	0.002898

Direction-specific scalar interpretation

- `Fw` : `H0` remains best. Every genuine PINN arm increases held-out MAE.
- `Bw` : `H1` improves test MAE by about `16.6%` relative to `H0` , but scalar gain alone is insufficient.
- `H3-Fw` : the Bauer-anchor test loss reaches approximately `9.01` , versus approximately `0.00455` on `Bw` . The frozen anchor is therefore not stable enough across directions to retain at the tested weight.
- the program winner remains the accepted `periodic_gru_sequence_bw` ; the program registry is not changed.

Common-Split Curve-Payload Evaluation

A bounded `CVP 1.2` diagnostic replayed

- the eight canonical Phase 2 checkpoints;
- the accepted non-windowed periodic harmonic MLP for `Fw` and `Bw` ;
- the accepted time-windowed periodic GRU for `Fw` and `Bw` ;
- exactly `97` common held-out curves per direction;
- full-resolution source curves for metrics;
- reduced payload samples only for repository-size control.

This is a Phase 2 closeout diagnostic, not the heavy official `TE Curve Verification Pipeline` refresh and not an automatic registry promotion. MAE and closure are in degrees, MPE and amplitude error are percentages, and phase error is in degrees.

Rank	Candidate	Surface	MAE	MPE	Amp. Err.	Phase Err.	Closure
1	accepted periodic harmonic MLP	`Fw`	0.001694	3.439	15.380	13.592	0.000881
2	accepted periodic GRU	`Fw`	0.001618	3.278	19.237	12.167	0.000892
3	accepted periodic harmonic MLP	`Bw`	0.001912	3.581	21.132	17.127	0.000861
4	accepted periodic GRU	`Bw`	0.001837	3.494	31.498	15.321	0.001313
5	Phase 2 `H0`	`Fw`	0.001998	4.174	39.328	21.029	0.000872
6	Phase 2 `H1`	`Fw`	0.002232	4.734	43.016	19.899	0.000857
7	Phase 2 `H2`	`Fw`	0.002102	4.407	49.491	20.335	0.000938
8	Phase 2 `H1`	`Bw`	0.002186	4.234	52.097	24.208	0.000865
9	Phase 2 `H3`	`Bw`	0.002386	4.710	56.022	25.209	0.000864

Rank	Candidate	Surface	MAE	MPE	Amp. Err.	Phase Err.	Closure
10	Phase 2 `H0`	`Bw`	0.002494	5.021	48.979	33.574	0.000868
11	Phase 2 `H2`	`Bw`	0.002414	4.802	60.967	25.537	0.000869
12	Phase 2 `H3`	`Fw`	0.002651	5.644	69.329	27.110	0.000957

The four accepted baselines occupy the first four composite diagnostic positions. No Phase 2 candidate displaces either the time-windowed or non-windowed accepted reference on its valid surface.

The scalar test loop and the full-curve replay use the same held-out conditions but different angular sampling. The scalar metrics use the training data-module stride and distributed point cap; the closeout diagnostic uses every source-curve point. Their absolute MAE values are therefore not expected to be identical.

Dominant And Selected Harmonic Evidence

Order `1` is dominant on both surfaces, with mean truth amplitude close to `0.0172 deg`. The next strongest mean amplitudes are:

- `Fw`: orders `39`, `78`, `40`, and `3`;
- `Bw`: orders `156`, `78`, `3`, and `162`.

Relative to `H0`, `H1-Bw` produces a real but incomplete improvement:

Order	H0 Amp. Error [%]	H1 Amp. Error [%]	H0 Phase Error [deg	H1 Phase Error [de
1	3.409	0.777	0.476	0.308
3	24.878	15.860	17.565	16.569
39	32.804	18.625	27.104	29.702
40	61.137	23.522	48.798	16.044
78	29.109	55.215	35.880	15.069
156	84.639	137.757	56.514	36.276
240	66.010	98.020	33.241	26.228

This shows why scalar-only acceptance would be misleading. The oscillator prior helps the dominant order and several low or intermediate orders, but it redistributes error into other physically relevant harmonics.

On `Fw`, `H1` slightly improves mean phase error but worsens mean amplitude error and raw curve error. `H2` does not recover a consistent advantage. `H3` becomes the weakest Phase 2 candidate on `Fw`.

Physics Residual Evidence

The deterministic validator proves

- an admissible sine/cosine component has maximum normalized residual near `1.19e-7`;
- a deliberately inadmissible component has residual mean square near `1.57`;
- enabled physics terms backpropagate nonzero gradients;
- explicit and implicit inference paths remain finite.

The real campaign proves the residual participates in optimization, but a smaller residual does not imply a better TE curve. Representative held-out physics diagnostics are:

Arm	Surface	Oscillator Loss	Periodic Value Lo	Periodic Slope Los	Bauer Anchor Loss
`H1`	`Fw`	0.02585	0.000832	0.90053	0

Arm	Surface	Oscillator Loss	Periodic Value Loss	Periodic Slope Loss	Bauer Anchor Loss
`H1`	`Bw`	0.02576	0.000582	5.81174	0
`H2`	`Fw`	0.01861	0.002293	0.001295	0
`H2`	`Bw`	0.02517	0.000065	0.001812	0
`H3`	`Fw`	0.02207	0.005010	0.001813	9.01045
`H3`	`Bw`	0.01794	0.001408	0.000636	0.004550

The periodic slope term works as intended in `H2` and `H3`, reducing the unweighted diagnostic by several orders of magnitude relative to `H1`. However, the offline closure mismatch was already approximately `0.0009 deg` for all candidates. Tightening the mathematical boundary does not materially improve the measured curve surface.

Exit-Gate Decision

The Phase 2 exit rule was

> Promote only constraints that improve held-out harmonic fidelity without > degrading raw error, offset behavior, or direction-specific continuity.

Decision: **Phase 2 complete; no physical constraint promoted.**

Rationale

1. `Fw` retains the Fourier control over every genuine PINN arm on raw and

   composite curve evidence.
2. `H1-Bw` is the strongest physics signal, but its amplitude improvement is

   not consistent across selected orders.
3. periodic closure reduces its own residual but does not create a

   compensating curve-first gain.
4. the Bauer anchor is directionally unstable at the tested weight and is

   rejected as a default training term.
5. accepted GRU and harmonic-MLP baselines remain stronger on the common

   held-out surface. No repeat-seed or broad physics-weight escalation is justified for a non-passing arm. The negative screen is the stopping rule that prevents spending additional training budget on a formulation that failed its initial multi-index gate.

Retained Engineering Value

The phase still delivers reusable infrastructure

- a differentiable mechanism-grouped harmonic-head PINN;
- higher-order angular automatic differentiation;
- bounded target-free collocation;
- separate oscillator, periodic-value, periodic-slope, and analytical-anchor

  logging;
- exact common-split support in training and CVP playback;
- immutable campaign checkpoint snapshots;
- full per-order harmonic diagnostic export;
- a local and `-Remote` campaign package;
- an inspectable inference path compatible with later TwinCAT simplification.

  Phase 3 may reuse the condition encoder, explicit intermediate quantities, common split, controls, and reporting surface. It must not inherit nonzero Phase 2 physics weights by default.

Registry And Program Status

- The campaign winner artifact remains `H0-Fw`, explicitly labeled scalar.
- The current program-best registry remains

  `te_periodic_gru_sequence_bw`.
- No official CVP recommendation changes.
- `H1-Bw` is retained as exploratory evidence, not an accepted model.
- The general full-PINN roadmap advances to Phase 3:

  quasi-static compliance and elastic-offset PINNs.
- The paper-faithful MMT branch remains deferred and does not block Phase 3.

Artifact Inventory

Canonical campaign

- `output/training_campaigns/2026-07-26-14-03-44_phase2_harmonic_kinematic_pinn_runtime_bounded_restart_2026_07_26/`
- `campaign_leaderboard.yaml`
- `campaign_best_run.yaml`
- `campaign_best_run.md`
- `campaign_execution_report.md`
- `candidate_registries/`

  Curve-payload evidence:
- `output/validation_checks/phase2_harmonic_kinematic_pinn_curve_payload_diagnostics/2026-07-26-16-49-48__track2c_curve_payload_diagnostics/`
- `candidate_payload_diagnostics.csv`
- `curve_payload_diagnostics.csv`
- `harmonic_payload_diagnostics.csv`
- `curve_payload_samples.jsonl`
- `track2_curve_payload_diagnostics_summary.yaml`

  Documentation:
- `doc/reports/analysis/model_development_waves/wave_5_2/harmonic_kinematic_pinn/[2026-07-26]/phase2_harmonic_kinematic_pinn_model_report.md`
- `doc/reports/analysis/model_development_waves/wave_5_2/harmonic_kinematic_pinn/curve_payload_diagnostics/[2026-07-26]/track2_curve_payload_diagnostics_report.md`
- `doc/reports/campaign_plans/model_development_waves/wave_5_2/harmonic_kinematic_pinn/2026-07-25-20-44-23_phase2_harmonic_kinematic_pinn_campaign_plan_report.md`

Next Step

Prepare Phase 3 as a new isolated formulation:

- bounded signed torque-to-deflection compliance;
- temperature-conditioned stiffness;
- direction-specific elastic and backlash offsets;
- zero-torque intercept and monotonicity checks;
- explicit comparison against the accepted baselines and the Phase 2 Fourier
  control;
- no automatic reuse of the rejected Phase 2 physical losses.